ECG Arrhythmia Classification and Calibration Study

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Abstract

Electrocardiogram (ECG) analysis is widely used for detecting cardiac rhythm abnormalities. Machine learning methods have shown promise for automated arrhythmia detection, but reliable clinical deployment requires careful model evaluation, probability calibration, and robustness assessment.

This study develops a reproducible machine learning pipeline for heartbeat classification using the MIT-BIH Arrhythmia Database. ECG beats are segmented and represented using physiologically meaningful features derived from waveform morphology and RR interval timing. Baseline classifiers are trained using leakage-safe record-wise data splitting. Model performance is evaluated using balanced accuracy, ROC curves, and precision–recall analysis. Calibration of probabilistic predictions is assessed using Brier scores and reliability diagrams, with Platt scaling applied to improve ventricular arrhythmia probability estimates. A robustness experiment is also performed by introducing synthetic noise to simulate signal perturbations.

Results demonstrate strong discrimination for ventricular arrhythmias and improved probability calibration after post-hoc scaling. The study highlights the importance of evaluation methodology and calibration when applying machine learning to cardiovascular signal analysis.

1 Introduction

Cardiovascular disease remains a leading cause of mortality worldwide, and electrocardiography is a primary tool for assessing cardiac rhythm abnormalities. Arrhythmias such as ventricular ectopic beats and supraventricular abnormalities can often be detected through irregularities in ECG waveforms or variations in heartbeat timing.

Recent advances in machine learning have enabled automated approaches for analysing ECG signals and detecting arrhythmias. However, many machine learning studies focus primarily on classification accuracy without addressing critical aspects of clinical modelling such as patient-level data leakage, probability calibration, and model robustness.

This work presents a reproducible pipeline for ECG heartbeat classification using the MIT-BIH Arrhythmia Database. In addition to training baseline machine learning models, the study emphasises evaluation practices relevant to medical AI systems. These include record-wise dataset splitting, probabilistic calibration analysis, and robustness testing under simulated noise conditions.

2 Dataset

The experiments use the MIT-BIH Arrhythmia Database available through PhysioNet [Moody and Mark, 2001, Goldberger et al., 2000]. The dataset contains 48 annotated ECG recordings sampled at 360 Hz. Each recording includes expert-labelled heartbeat annotations.

Heartbeats were mapped to the AAMI classification scheme commonly used in arrhythmia research.

Class	Description
N	Normal beats
S	Supraventricular ectopic beats
V	Ventricular ectopic beats
F	Fusion beats
Q	Unknown or paced beats

Table 1: AAMI heartbeat classification scheme used in this study.

After segmentation and preprocessing, approximately 109,000 heartbeats were available for modelling.

To avoid patient-level data leakage, dataset splitting was performed at the record level rather than at the individual beat level. The resulting splits were:

- Training set: 33 records
- Validation set: 5 records
- Test set: 10 records

This ensures that beats from the same ECG recording do not appear in both training and evaluation sets.

3 Feature Engineering

Two categories of physiologically meaningful features were extracted from each heartbeat.

3.1 Timing Features

Heartbeat timing characteristics were derived from RR intervals between consecutive beats. These features capture rhythm irregularity, which is a key indicator of many arrhythmias.

Extracted features include:

- Previous RR interval
- Next RR interval
- Heart rate derived from RR intervals
- RR interval ratio

3.2 Morphological Features

Additional features describe the shape and dynamics of the ECG waveform around each beat.

These include:

- Mean amplitude
- Standard deviation
- Minimum and maximum amplitude
- Peak-to-peak amplitude
- Signal energy

- Absolute mean amplitude
- Derivative slope statistics

These features capture morphological differences in ECG waveforms such as QRS complex variations.

4 Models

Two baseline classifiers were implemented.

4.1 Logistic Regression

Logistic regression provides an interpretable baseline for multi-class classification. Balanced class weighting was applied to address class imbalance in the dataset.

4.2 Random Forest

A random forest classifier with 300 decision trees was used to capture non-linear relationships between features. Class weights were also balanced to mitigate skewed class distributions.

5 Evaluation Metrics

Model performance was assessed using several metrics relevant to imbalanced classification problems.

- Balanced accuracy
- ROC-AUC (Receiver Operating Characteristic)
- Precision–Recall AUC
- Confusion matrices

Balanced accuracy was used as the primary metric due to the large imbalance between normal and arrhythmic beats.

6 Results

ROC curves for each heartbeat class are shown in Figure 1.

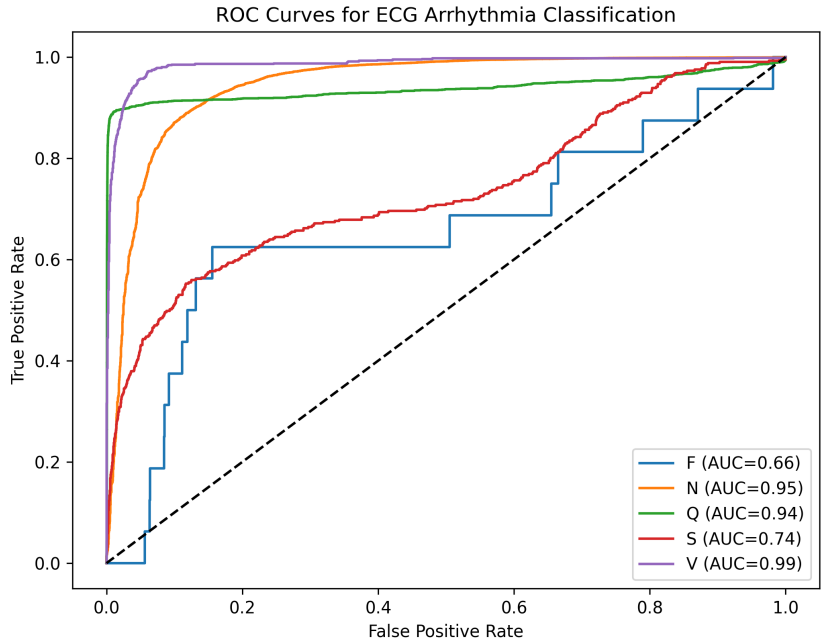

Figure 1: ROC curves for arrhythmia classification across heartbeat classes.

The ventricular arrhythmia class achieved strong discriminative performance with an ROC-AUC close to 0.99. Normal and unknown classes also demonstrated high separability, while rarer classes such as supraventricular and fusion beats were more challenging to distinguish.

7 Probability Calibration

For clinical prediction systems, reliable probability estimates are essential. Calibration was therefore evaluated using Brier scores and reliability diagrams.

Figure 2 shows the calibration curve for ventricular arrhythmia predictions.

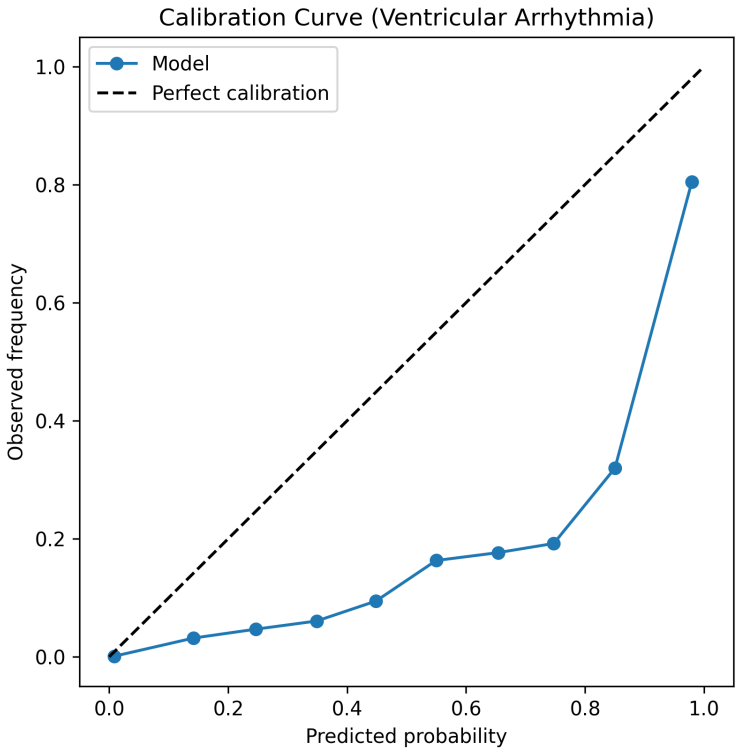

Figure 2: Calibration curve for ventricular arrhythmia predictions.

The baseline model showed a tendency to overestimate the probability of ventricular arrhythmias. To address this, Platt scaling was applied using a validation dataset.

The Brier score improved from:

$$0.0222 \rightarrow 0.0179$$

after calibration.

Figure 3 shows the improvement after applying Platt scaling.

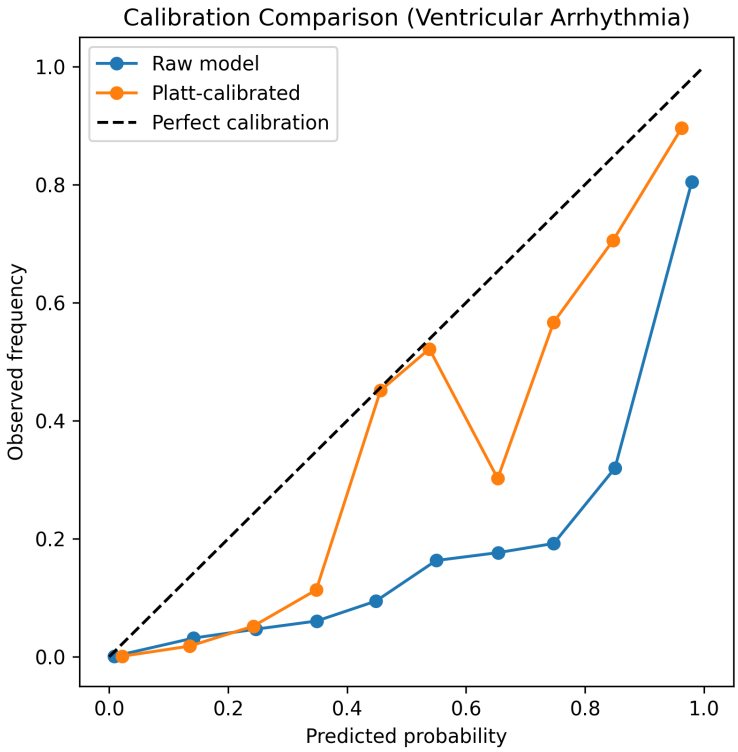

Figure 3: Comparison of raw and calibrated probability estimates.

8 Feature Importance

Feature importance analysis was conducted to identify the most influential predictors of arrhythmia classification.

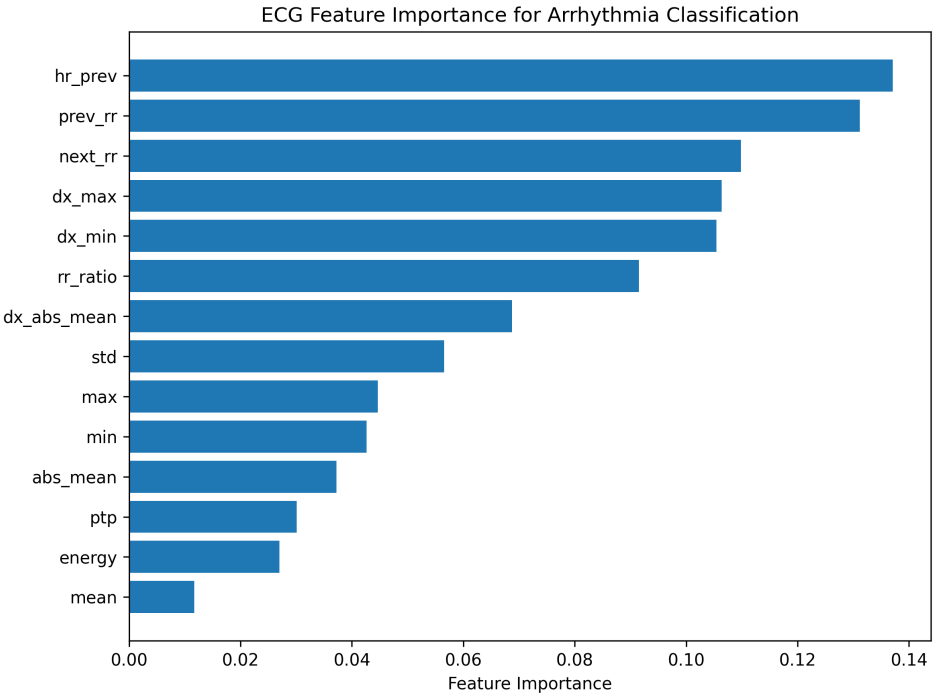

Figure 4: Feature importance for arrhythmia classification.

The most informative predictors were RR interval–based timing features, including previous heart rate and adjacent RR intervals. These findings align with known electrophysiological indicators of rhythm irregularity.

9 Robustness Analysis

To evaluate model stability under realistic signal perturbations, Gaussian noise was introduced to ECG-derived features.

Balanced accuracy decreased from:

$$0.67 \rightarrow 0.62$$

under noisy conditions.

This moderate performance degradation suggests the model retains reasonable stability when exposed to feature-level noise.

10 Discussion

The results demonstrate that classical machine learning models can effectively detect ventricular arrhythmias using features derived from ECG waveforms and heartbeat timing.

However, the study also highlights several methodological considerations important for clinical machine learning:

- Avoiding patient-level data leakage
- Evaluating probability calibration
- Interpreting physiologically meaningful features
- Assessing robustness to signal perturbations

Future work could explore deep learning models trained directly on raw ECG waveforms, cross-dataset validation, and patient-level risk aggregation.

11 Conclusion

This study presented a reproducible machine learning pipeline for ECG arrhythmia classification using the MIT-BIH dataset. The pipeline emphasised robust evaluation practices including leakage-safe dataset splitting, probability calibration, and robustness testing.

Results showed strong discrimination for ventricular arrhythmias and improved probability reliability following calibration. The study demonstrates the importance of combining machine learning modelling with rigorous evaluation methodology in cardiovascular AI research.

References

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